Unit 6 Lesson 2: Writing CSV Files

Quick Reference Guide

📦 Import

```
import csv
```

✏️ Create New CSV File (Write Mode)

```
data = [
    {'customer_id': 'C001', 'name': 'Acme Corp', 'revenue': 45000},
    {'customer_id': 'C002', 'name': 'TechStart Inc', 'revenue': 32000}
]

with open("customers.csv", "w", newline="") as file:
    fieldnames = ['customer_id', 'name', 'revenue']
    writer = csv.DictWriter(file, fieldnames=fieldnames)

    writer.writeheader()     # Write column names
    writer.writerows(data)   # Write all data
```

Result file:

```
customer_id,name,revenue
C001,Acme Corp,45000
C002,TechStart Inc,32000
```

➕ Append to Existing File

```
new_customer = {'customer_id': 'C003', 'name': 'Global Ltd', 'revenue': 28000}

with open("customers.csv", "a", newline="") as file:
    fieldnames = ['customer_id', 'name', 'revenue']
    writer = csv.DictWriter(file, fieldnames=fieldnames)

    writer.writerow(new_customer)  # NO header!
```

⚠️ Key Rules

Rule	Why It Matters
Always use `newline=""`	Prevents extra blank lines on Windows
"w" mode overwrites	Existing data is deleted!
"a" mode appends	Adds to end, keeps existing data
No header when appending	Header is already in the file

📝 writerow vs writerows

```
# writerow() - One record at a time
writer.writerow({'customer_id': 'C001', 'name': 'Acme Corp', 'revenue': 45000})

# writerows() - Multiple records at once
writer.writerows(list_of_customers)
```

🔚 Read-Modify-Write Pattern

```
import csv

# 1. READ - Load existing data
with open("inventory.csv", "r") as file:
    products = list(csv.DictReader(file))

# 2. MODIFY - Change the data
for product in products:
    # Apply 10% price increase
    product['price'] = float(product['price']) * 1.10

# 3. WRITE - Save changes back
with open("inventory.csv", "w", newline="") as file:
    fieldnames = ['product_id', 'name', 'price', 'quantity', 'category']
    writer = csv.DictWriter(file, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(products)
```

📁 File Modes Summary

Mode	Creates?	Existing Data	Use For
`"w"`	Yes	**Deleted**	New reports, complete exports
`"a"`	Yes	**Kept**	Adding transactions, new records
`"r"`	No	Read only	Loading data

🔑 fieldnames

The `fieldnames` list controls:

1. **Which keys** are written from each dictionary
2. **Column order** in the CSV file

```
fieldnames = ['employee_id', 'name', 'department', 'salary']
# Columns will appear in this exact order
```

❌ Common Mistakes

Forgetting newline=""

```
# ❌ Wrong
with open("sales.csv", "w") as file:

# ✅ Correct
with open("sales.csv", "w", newline="") as file:
```

Writing header when appending

```
# ❌ Wrong - header appears multiple times!
with open("transactions.csv", "a", newline="") as file:
    writer.writeheader()  # DON'T do this!
    writer.writerow(record)

# ✅ Correct
with open("transactions.csv", "a", newline="") as file:
    writer.writerow(record)  # Just the data
```

Using "w" when you meant "a"

```
# ❌ DANGER - deletes all existing data!
with open("customers.csv", "w", newline="") as file:

# ✅ Safe - adds to existing data
with open("customers.csv", "a", newline="") as file:
```

💼 Business Use Cases

Sales Reports

```
sales = [
    {'month': 'January', 'product': 'Laptops', 'units_sold': 42, 'revenue': 50400},
    {'month': 'January', 'product': 'Monitors', 'units_sold': 67, 'revenue': 20100}
]
```

Customer Management

```
customers = [
    {'customer_id': 'C001', 'name': 'Acme Corp', 'email': 'contact@acme.com'},
    {'customer_id': 'C002', 'name': 'TechStart Inc', 'email': 'info@techstart.com'}
]
```

Inventory Updates

```
inventory = [
    {'product_id': 'P001', 'name': 'Laptop', 'price': 1200, 'quantity': 45},
    {'product_id': 'P002', 'name': 'Monitor', 'price': 300, 'quantity': 120}
]
```

Employee Records

```
employees = [
    {'employee_id': 'E001', 'name': 'John Smith', 'department': 'Sales', 'salary': 65000},
    {'employee_id': 'E002', 'name': 'Sarah Johnson', 'department': 'Marketing', 'salary': 70000
]
```